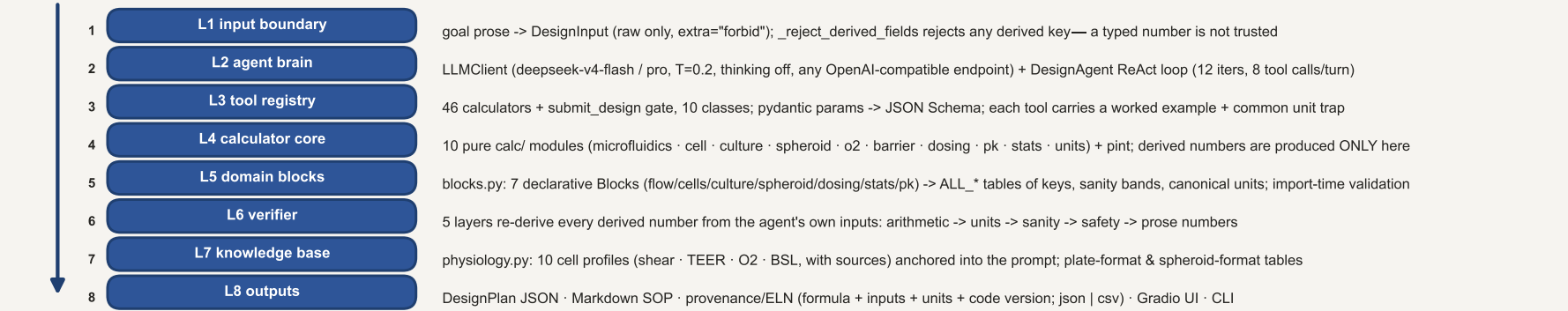
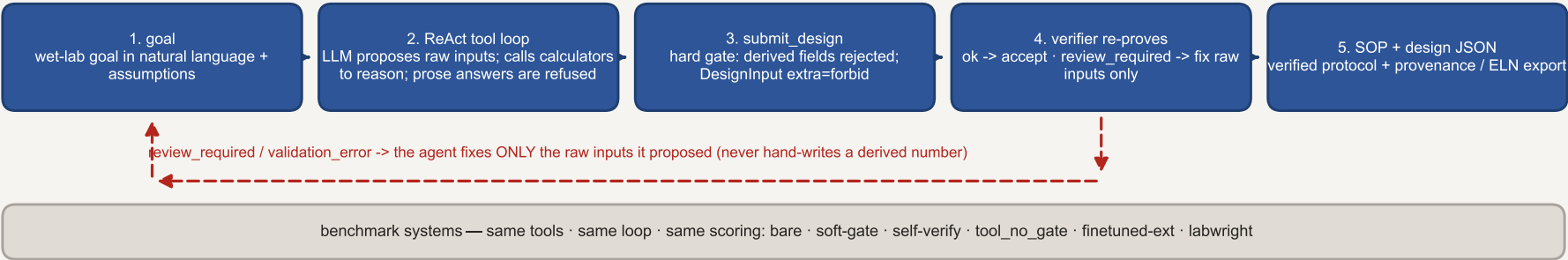


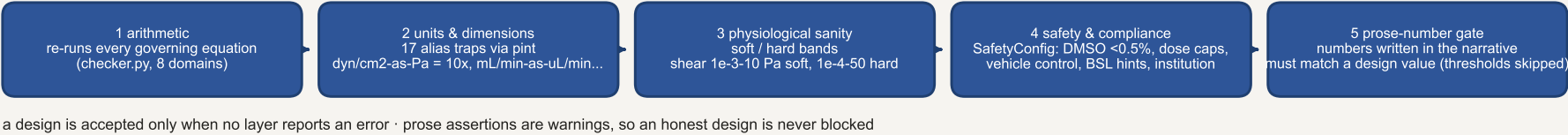
(a) layered architecture — the LLM proposes raw inputs; deterministic calculators compute; the verifier re-proves



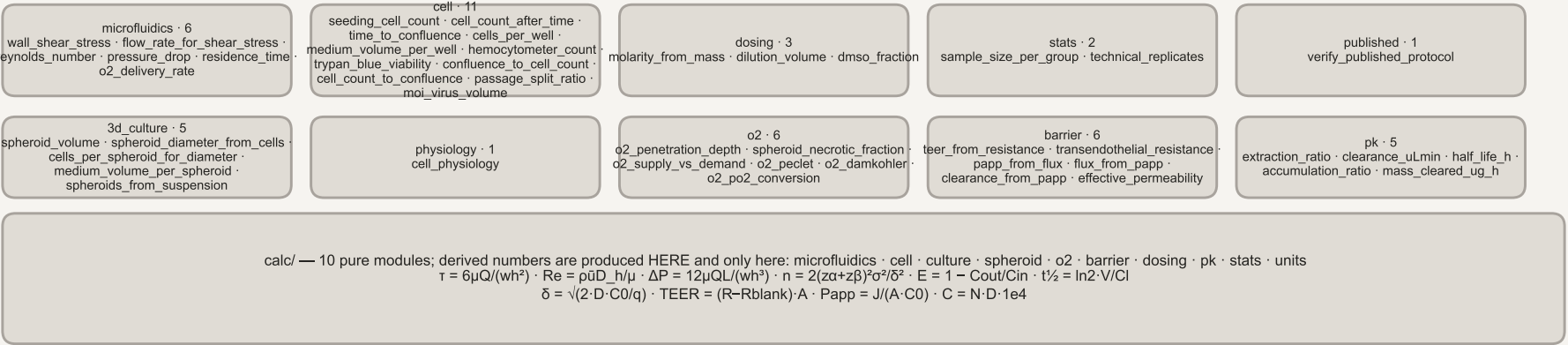
(b) bounded agentic workflow — one goal in, a design whose every number was computed by calc/ and re-proved by verify/ comes out



(c) verifier — re-derives every derived number from the agent's own raw inputs



(d) calculator toolbox — 46 tools in 10 classes (pure functions · pydantic-schema'd · pint units · worked example + unit traps)



every derived field pinned to one canonical unit by pint (dyn = 1e-5 N); a new domain = one calc/ module + one register_tool call — not a fork

(e) internal components · benchmark systems · honest boundary
the agent & brain

DesignAgent — ReAct loop; ends only on submit_design; prose refused; on review_required fixes raw inputs only
LLMClient — OpenAI-compatible; deepseek-v4-flash/pro; T=0.2; thinking off; retry 3x backoff; max_tokens 8192
tool_no_gate ablation — same calculators & loop, verifier off; isolates what the verification layer adds

cell_physiology — read-only lookup over the 10-profile physiology registry
verify_published_protocol — same calculators run backwards over a paper's own inputs (consistent / discrepancy / ...)
UI (Gradio two-axis badge) · CLI · Colab — all entry points into the same verified pipeline

fine-tuned extractor (single-pass, not an agent)

Qwen2.5-1.5B-Instruct + LoRA (r=16) tuned on goal -> raw-inputs; goes through the SAME gate (reject derived -> DesignInput -> build_design -> verify)

committed eval: JSON parse 1.0 · extract->verify 0.9976 vs untuned 0.40

the agent and the extractor are the two independent paths that produce raw inputs; both feed the identical calculators and verifier

gold sets: 24-reading · 15-blind (8 cold + 5 prompt-backed + 2 scenario) · 15-3D-spheroid · 14-plate-culture · 14-perfused-PK

metrics: usable · self-consistent · hallucination · failure reason · unit-misread · target-selection · cold-split

honest boundary & committed results

the verifier checks arithmetic & internal consistency, not target selection — the UI marks both (green = arithmetic verified / amber = targets model-proposed)
reading set (24): usable 88% flash / 100% pro; hallucination 0.125 / 0.000
5-seed Wilson 0.925 [0.864, 0.960] / 0.958 [0.906, 0.982]
blind set (15): usable 40% / 47% (cold-only 38%); hallucination 0.000 (attack-tested) · self-consistency 100%

hallucination == 0.000 is attack-tested (prose refusal · derived-field rejection · tamper detection · prose assertions), not "zero by construction"